

Fig. 3: An actual-size, single-side PCB for bedwetting alarm

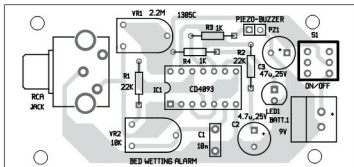


Fig. 4: Component layout for the PCB

PARTS LIST

Semiconductors:

- IC1 - CD4093 NAND gate
LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- | | |
|-------|------------------------|
| R1-R2 | - 22k Ω |
| R3-R4 | - 1k Ω |
| VR1 | - 2.2M Ω preset |
| VR2 | - 10 k Ω preset |

Capacitors:

- | | |
|----|---------------------------------|
| C1 | - 10nF ceramic |
| C2 | - 4.7 μ F, 25V electrolytic |
| C3 | - 47 μ F, 25V electrolytic |

Miscellaneous:

- BATT.1 - 9V battery
S1 - On/off switch
PZ1 - Piezo buzzer

tween MSA and MSB terminals. This provides a low level at the input of

N1 that triggers the oscillator and the buzzer turns on.

Construction and testing

An actual-size, single-side PCB for bedwetting alarm is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on a PCB, enclose it in a suitable case that should have an opening for the RCA socket, on/off switch, LED1 and buzzer as shown in Fig. 1.

The gap between the MSA and MSB should not be more than 50 mm. Use any twin-core screened cable for interconnection between the moisture

sensor and the audio alarm generator. Note that it is possible to replace the sensor elements (MSA and MSB) of moisture sensor MS1 by other thin conductive fasteners or foils.

To test the circuit, trigger switch S1 and check various voltages as shown in the table. Status of LED1 indicates the availability of 9 volts, which can also be verified at TP1 with respect to TP0. TP2 goes low when the bed sheet between MSA and MSB is wet, triggering the oscillator, which can be verified at TP3. ●

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